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1- INTRODUCTION

1-1 Overview

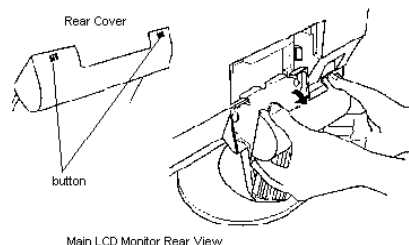
This procedure will describe how to adjust the LCD display monitor to match the camera. Once the LCD display monitor is re-calibrated, it is essential to check and possibly re-calibrate the camera.

1-2 Camera Vendor Participation

This procedure should be performed with the Camera Vendor field engineer, present, should any camera adjustment be necessary. It is also recommended that you have one of the Customer's filming specialists available to review the film/LCD display monitor conformance.

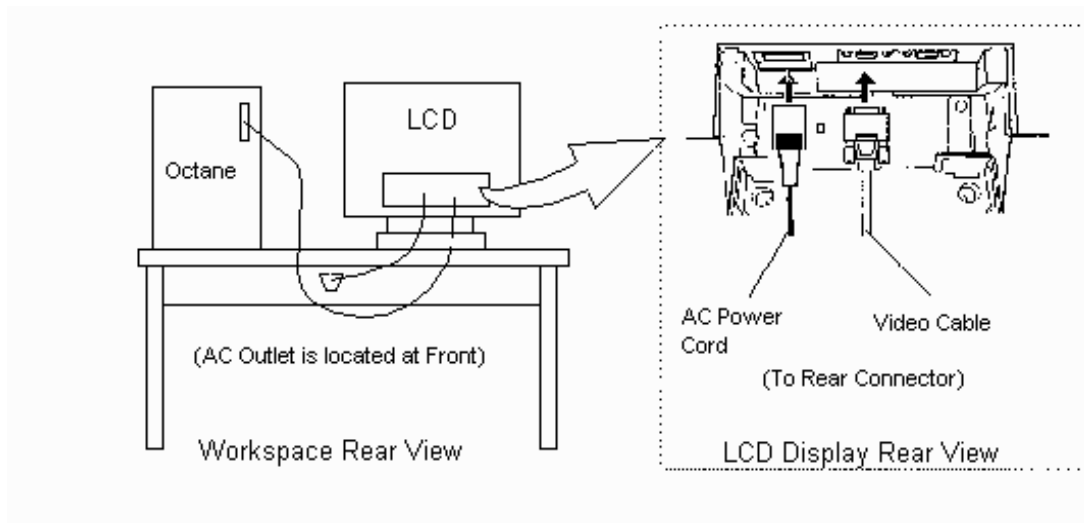
2 – EIZO MONITOR INSTALLATION

1. Shutdown Signa and perform a safety lockout/tagout of the Operator Workspace.
2. Remove the new LCD monitor from its box and its packing and set the Eizo L660 LCD color monitor on the workspace desktop and remove its rear cover. See Illustration 2-1



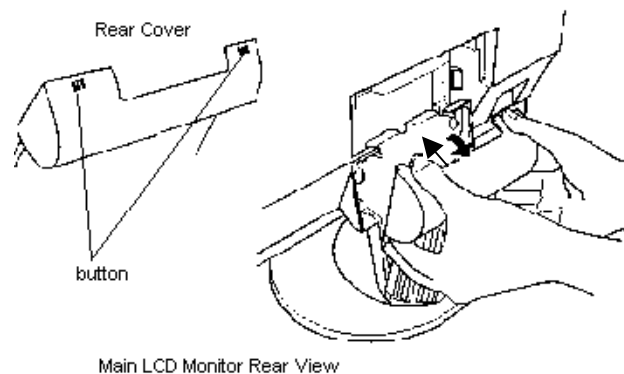
REMOVING BASE COVER
ILLUSTRATION 2-1

1. Connect the Video and power cables. Route these cables through the base. See Illustration 2-2.



VIDEO AND POWER CABLE CONNECTIONS
ILLUSTRATION 2-2

2. Re-attach the cover to the back of the LCD. See Illustration 2-3

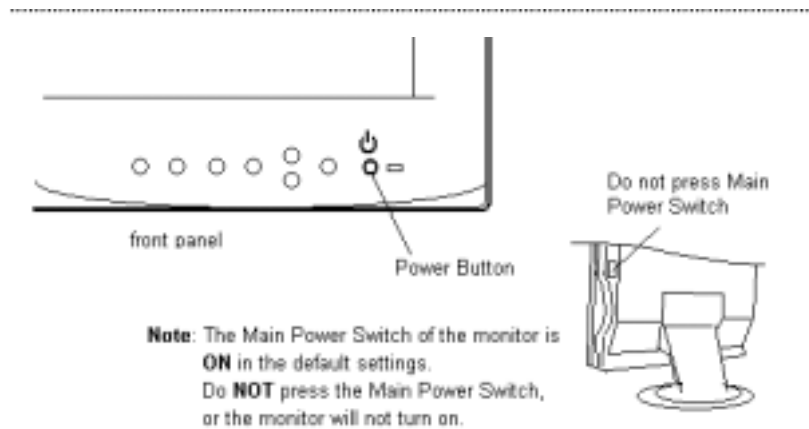


REPLACING THE BASE COVER
ILLUSTRATION 2-3

3. Remove the Safety Lockout of the Operators Workspace and restore power to the system.
4. Verify that main LCD monitor power is ON. If not, press the power button on the front panel to turn the main LCD monitor ON. See Illustration 2-4.
5. The monitor should be positioned no closer than 16 inches and no further away then 28 inches from your eyes. The optimal distance is 24 inches for either of the monitors.

Note

Allow the monitor to warm-up for 20 minutes before performing any adjustments.



POWER BUTTON AND POWER SWITCH LOCATION
ILLUSTRATION 2-4

6. Boot up Signa

7. Password: **adw2.0**

3- VERIFY AMBIENT LIGHTING CONDITIONS

LCD's are extremely susceptible to Viewing Angle. All adjustments and image comparisons should be done from exactly the same viewing angle in all directions. Sit directly in front of the monitor at all times.

In the review area and operator workspace area, verify that the ambient lighting conditions are adjusted to a minimum level. In the operator workspace area, there should be only sufficient light for safely operating the system.

In the review area and operator workspace area, verify that light-boxes are not emitting light, or are properly masked, when not displaying film. This will be a source of excessive glare

In both review area and operator workspace area, verify that there is no source of glare for reviewing films or setting up the images for film. For example, windows should not allow direct light. (blinds should be closed).

Note that both, the operator workspace area and the review area artificial lighting type should be of the same type.

4- SETTING LCD MONITOR FREQUENCY RATE AND RESOLUTION

Failing to do this section will result in Image Ghosting and unnecessary Troubleshooting.

4-1- Setting the Line Rate and Resolution



THIS PROCEDURE NEEDS TO BE DONE EVERY TIME SOFTWARE HAS BEEN RELOADED. LINE RATE IS NOT CAPTURED ON THE SAVEINFO MOD AND IS NOT CAPTURED IN THE HEALTH PAGE STATUS OF THE SYSTEM.

Monitor "Line Rate" or "Refresh Rate", is referred to as "Frame Rate" and Swap Rate", on SGI systems. For simplicity, the term "Line Rate" will be used in this document.

Note:

Failure to set the Line Rate to 60Hz for ALL LCD Host monitors will result In Image Ghosting and customer complaints. This procedure needs to be done EVERY TIME software has been reloaded. Line Rate is NOT captured on the SaveInfo MOD or in the Health Page Status of the system.

Note:

UNIX/IRIX is case sensitive. Use the commands exactly as shown as "root owner".

1. From the Desktop click on the **C-shell tool**
2. Switch to root owner. Type: **su root** [enter]; (Enter password)
3. Type: **chkconfig fixed72hz off** [enter]

Note:

If you get the following error you may have miss-typed the command. Re-type the command. If it appears the second time, IGNORE IT. Proceed with the step 4.

chkconfig: cannot set flag "fixed72hz": No such file or directory
use "chkconfig -f fixed72hz off" to create and set the flag

4. Type: **/usr/gfx/setmon -x -w 1280x1024_60**
5. Type: **exit** [enter] to exit root
6. Type: **exit** [enter] to exit the C-shell
7. Select **System Shutdown**, (OK to shutdown)
8. Select **Restart**
9. **Choose Signa icon**, (Enter password)
10. Proceed to Section 5 : Signa Host Gamma Setup – EIZO L660

Note:

To check line rate values, refer to Troubleshooting section at the end of this procedure.

5- SIGNA HOST GAMMA SETUP- EIZO L660

The GAMMA value is modified to optimize the contrast level of the image mid-tones to more closely represent the same contrast that is filmed.

Note:

At time of System Installation:

- As of July 28, 2003 all NEW LCD monitors will have a Gamma Software CDROM disk INCLUDED in the box, (GE Part # 2376768) which should be used for software versions based on 9.0 and below. Use this CDROM disk and the instructions it contains to setup gamma for those software versions.

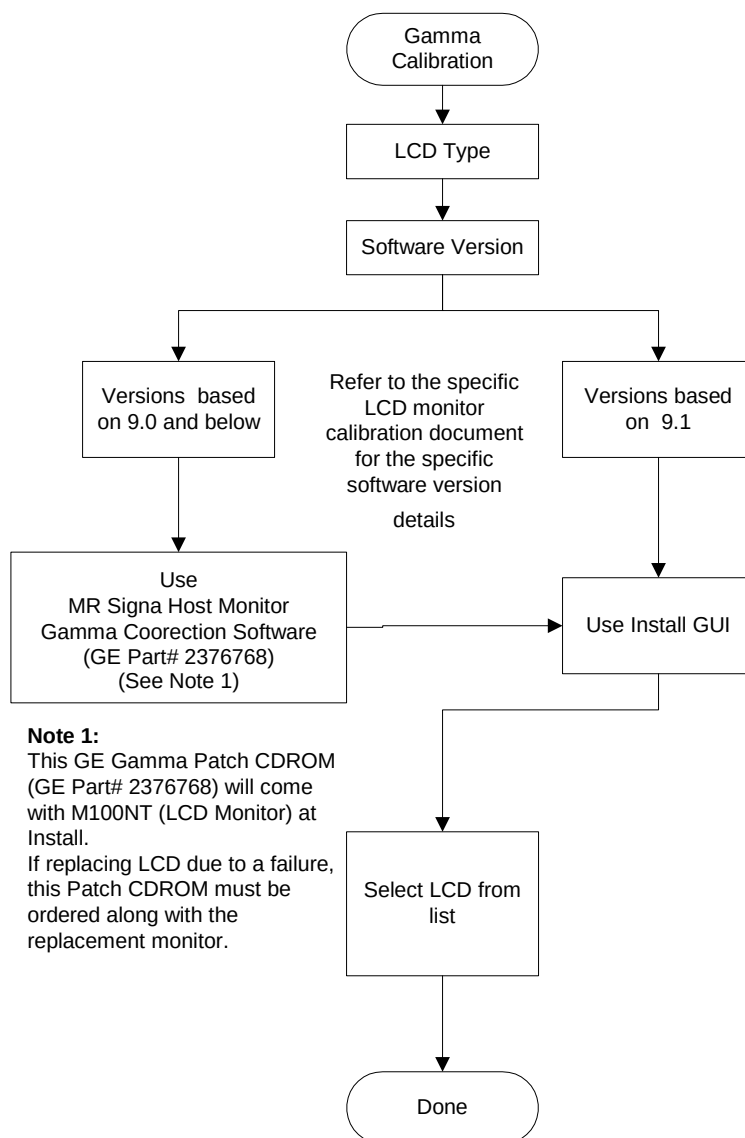
Upon replacement of a failed monitor:

Make sure you also order:

GE Part # 2376768. This CDROM disk is labeled “MR Signa® Host Monitor Gamma Correction Software” contains the most up-to date gamma files needed to setup gamma. The software on the CDROM is required to install a replacement monitor. Follow the installation instructions included with the CDROM disk.

WARNING!

MR SIGNA® HOST MONITOR GAMMA CORRECTION SOFTWARE (GE PART # 2376768) SUPERCEDES ALL OTHER GAMMA FILES THAT MAY BE ON VERSION 14 AND 15 LX/8X SERVICE METHODS CDROM.



PROCESS FLOW CHART
ILLUSTRATION 5-1

5-1 Gamma Configuration for all SGI Computers

Select the section to use based on your Computer type and Software version, using the table below:

GAMMA PROCEDURES BY SECTION

TABLE 5-1

Computer Type:	Software Version:	Gamma Setting:
All SGI computers	ALL Software versions 9.0 and below. Setting Gamma <u>with</u> the CDROM disk that came with the monitor.	See Section 5-1-1
Octane 1 and Octane 2	9.1, HFO2, HF03 Setting Gamma on monitors <u>without</u> a CDROM Disk	See Section 5-1-2

5-1-1 Setting Gamma with the Patch CDROM – All Software versions 9.0 and below

- If you are Installing the EIZO L660 monitor for the first time a CDROM labeled: MR Signa[®] Host Monitor Gamma Correction Software is included in the Monitor box. The installation instructions are included with the CDROM disk and should be followed to load the gamma software on the system.
- If you are replacing a failed LCD Monitor, then a CDROM labeled: MR Signa[®] Host Monitor Gamma Correction Software, (GE Part # 2376768) should have been ordered along with the replacement monitor. The installation instructions are included with the CDROM disk and should be followed to load the gamma software on the system.
- This latest CDROM disk should be retained and re-used any time a reload of software occurs on the system.
- If you do not know the software version, type “getver” in a c-shell.

This option is used to configure an EIZO L660 LCD. To set gamma:

- **If you are replacing the EIZO L660 with another L660 and have not reloaded software:** No gamma changes are necessary. Proceed to Subsection 6 in this document.
- **If you are replacing the EIZO L660 with the another L660 and have reloaded software:** Proceed to STEP 1 below:
 1. Insert the “MR Signa[®] Host Monitor Gamma Correction” CDROM into the Host CDROM/DVDROM drive.
 2. Follow the instructions included with the CDROM to set the gamma for your monitor type.
 3. Proceed to section 6 in this document.

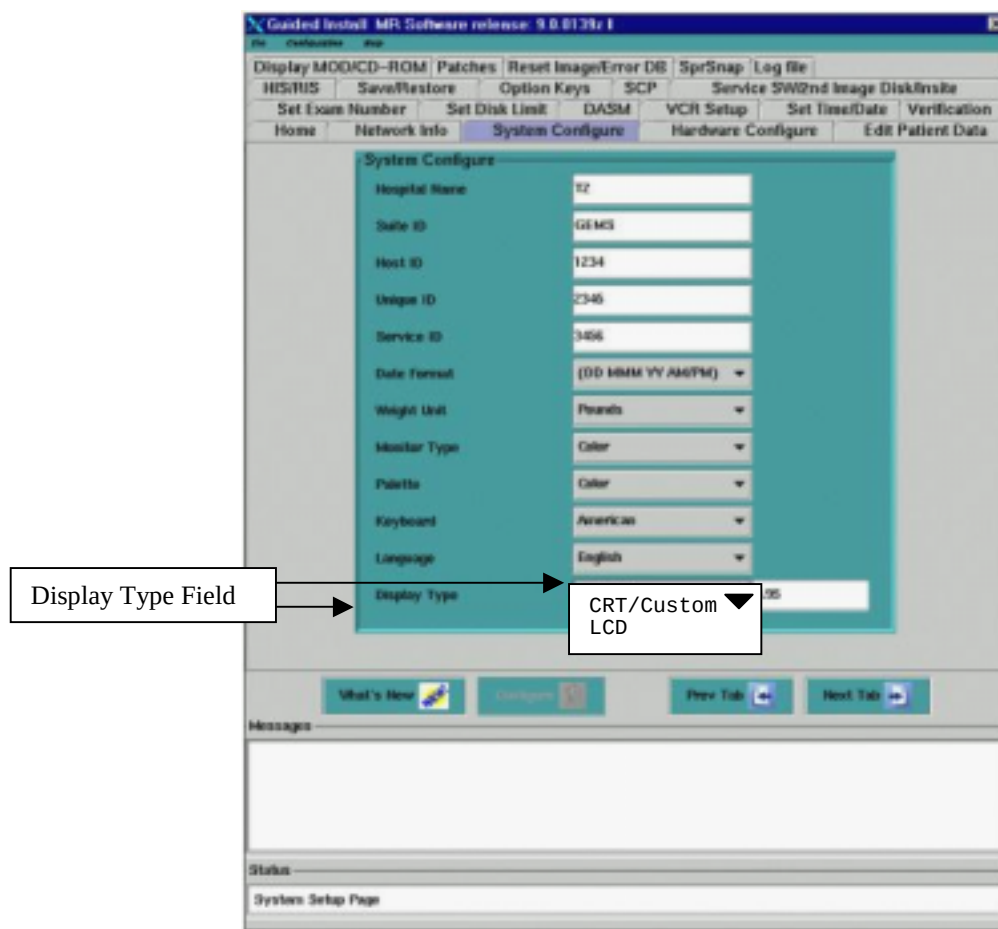
5-1-2 Setting Gamma - 9.1, HF02 and HFO3

This option is used to configure an EIZO L660 LCD. To set gamma:

- **If you are replacing the EIZO L660 with another L660 and have not reloaded software:** No gamma changes are necessary. Proceed to Subsection 6 in this document.
- **If you are replacing the EIZO L660 with the another L660 and have reloaded software:** Proceed to STEP 1 below:

1. Open Guided Install and select **System Configure** tab. Refer to illustration 5-1.

1. Look for the “**Display Type**” field. (See Illustration 5-1)



GUIDED INSTALL GUI “SYSTEM CONFIGURE” TAB- DISPLAY TYPE FIELD
ILLUSTRATION 5-1

Note:

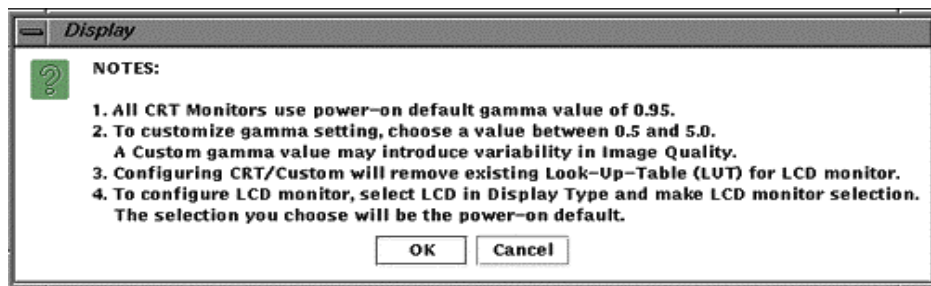
The “Display Type” Field has two choices:

- **LCD** – This should be used to configure Gamma for a specific LCD listed in the dropdown box. (Preferred method)
- **CRT/Custom** – This should be used to set a Custom LCD or CRT Gamma setting.

Note:

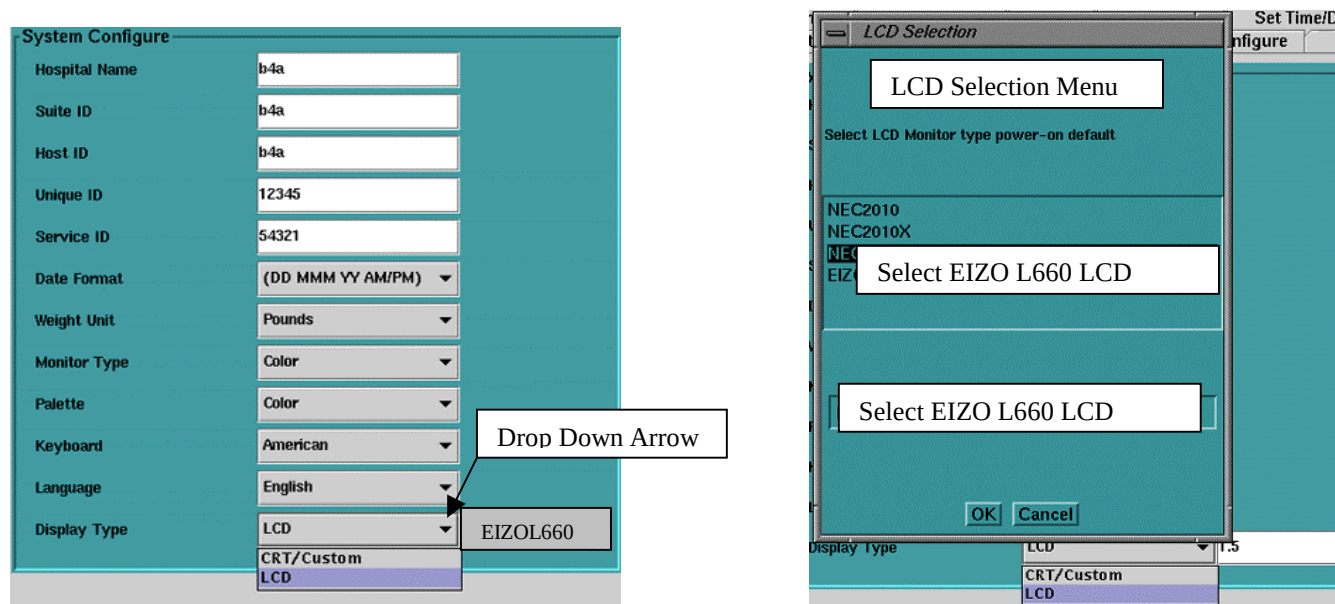
It may be necessary to “Toggle” between the “**CRT/Custom**” and “**LCD**” options in the Display Type Field to select “**LCD**”.

1. If “**CRT/Custom**” is displayed in the Field:
 - a. Click on the “Display Type” Field ▼.
 - b. Drag your mouse over **LCD**
 - c. A Popup Display message will appear, Select [OK] (See illustration 5-2)



MESSAGE BOX
ILLUSTRATION 5-2

- d. The Field should change to [LCD].
- e. Select **LCD** again. The following menu will appear:



MONITOR CHOICES
ILLUSTRATION 5-3

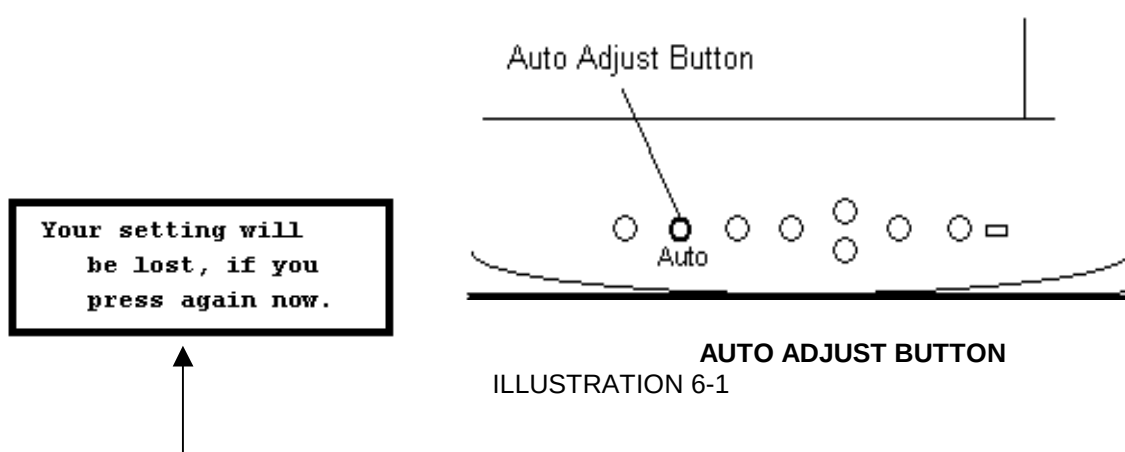
- a. **Scroll to the correct Display option (EIZOL660) and release the mouse button.** The display selected will appear in the box to the right.
- b. Click on the [Configure] Button on the Install GUI to save the changes. You may see a slight change in the screen intensity when “Configure” is complete.
- c. A Popup message box will appear, Select **OK**.
- d. Select **File** and **Quit** and **OK** from the Install/Reconfigure GUI Window.

- e. **Reboot the System.** A reboot is required to activate the changes. This completes the Gamma Calibration process.
- f. Proceed to Section 6- LCD Auto Adjustment and Screen Manager Operations.

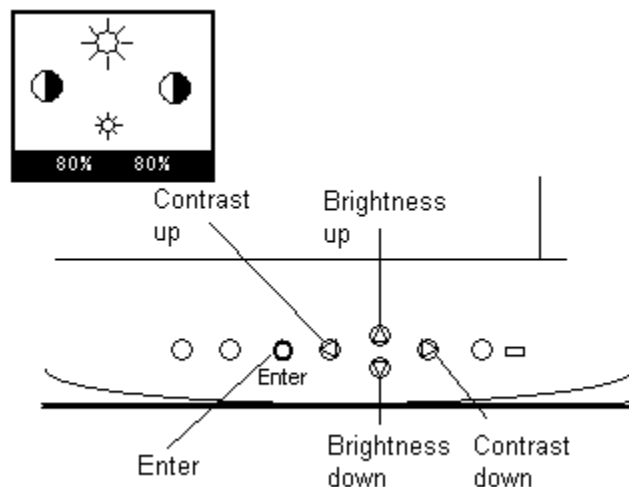
6- LCD AUTO ADJUSTMENT AND SCREEN MANAGER OPERATIONS

This section should get you familiar with the basic operation of the monitor controls. The intent of this section is to adjust the monitor to a "default" condition to proceed with calibrations. It also will allow you to become familiar with the button panels and the Screen Manger setup.

1. With the monitor ON, press the Auto Adjust Button.



2. While the message is on the screen (within 5 seconds), press the Auto Adjust Button again to automatically center the screen.
3. Press the Contrast up button. Set to 80%.
4. Press the Brightness up button. Set to 80%.



ADJUSTMENT
ILLUSTRATION 6-2

5. Press the Enter Button. All adjustments are now saved. When Auto Adjust button is pushed, the LCD will return to these settings. These settings will remain in effect even if power is removed, or system software is re-booted. Unless manually changed by the operator in the Screen Manager.
6. For further information, of the Screen Manager and button identification, refer to Appendix A. You may also review the EIZO User's Manual that came with the LCD Display.

The LCD Display Monitor is now ready for calibration.

Note

For further information, of the Screen Manager and button identification, refer to the Eizo L660X User's Manual that came with the LCD Display.

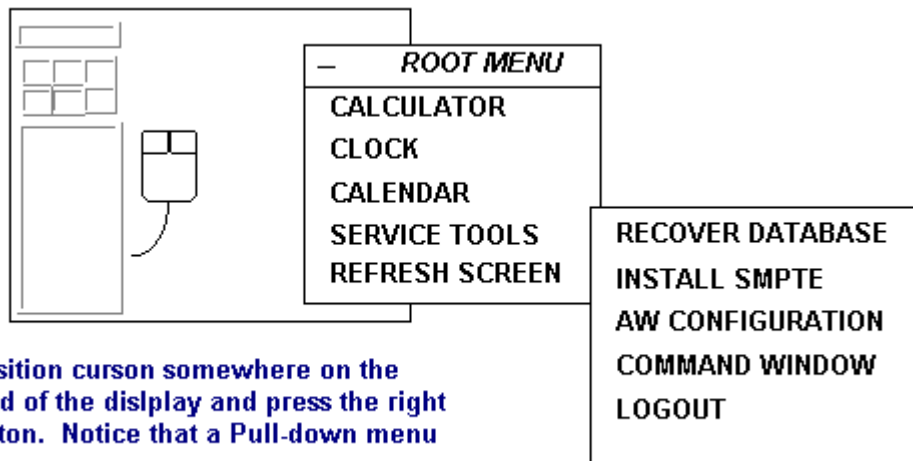
7- LCD CALIBRATION FOR OPTIMUM IMAGE VIEWING

The SMPTE (Society of Motion Picture and Television Engineers) test pattern is used to provide a standard image for calibrating the Window and Level of the display for an MR scanner.

7-1 Displaying the SMPTE Pattern

The Window and Level adjustments must now be set to ensure site to site uniformity in image appearance. The SMPTE test pattern is available on the Operator Workstation Host Computer after IRIX, (The operating system) has been booted, and after you have logged into the system.

1. Install and display the SMPTE pattern. Using the steps in illustration 7-1 Below :



Step 1. Position cursor somewhere on the background of the display and press the right mouse button. Notice that a Pull-down menu appears.

Step 2. Slide the mouse down to [Service Tools], and then over to [Install SMPTE]. This will load the SMPTE pattern into the database as Image 1000.

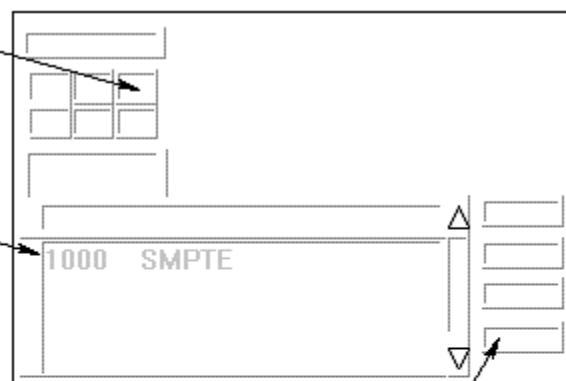
INSTALLING THE SMPTE PATTERN ILLUSTRATION 7-1

2. Displaying the SMPTE Pattern by following the steps in illustration 7-2 on the next page. Switch to the Browser and select the SMPTE pattern from the patient list and display it.
3. If necessary, use the FORMAT option to select Full Screen mode.

Step 1. Point to and click on the Display icon. Notice that the Browser comes up.

Step 2. After the Browser comes up, use the scroll bar on the right side of the display to find Image 1000 SMPTE.

Step 3. Point to and single-click on the SMPTE entry.

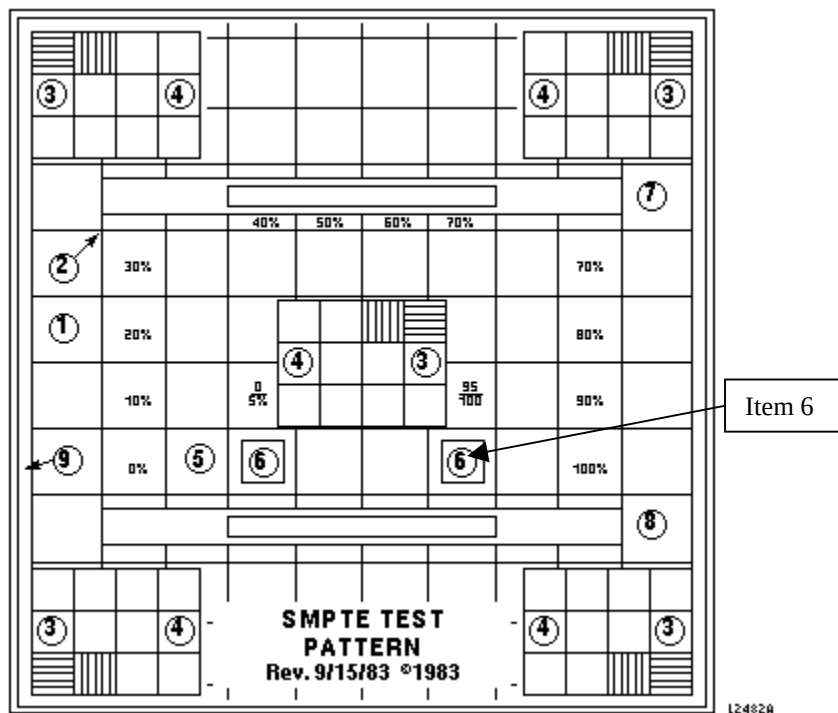


Step 4. Point to and single-click on the Full Viewer to display the SMPTE so that it is full screen size.

Step 5. To get back from displaying the SMPTE pattern, hit <ESC>.

INSTALLING THE SMPTE PATTERN ILLUSTRATION 7-2

7-2 Adjusting for proper Window and Level Setting



SAMPLE SMPTE TEST PATTERN
ILLUSTRATION 7-3

1. With the cursor inside the displayed image, hold down the middle mouse button and move the mouse in the horizontal plane. View the window control value at the base of the image and **set window control value to 100**.
2. With the cursor inside the displayed image, hold down the middle mouse button and move the mouse in the vertical plane. View the level control value at the base of the image and **set level control value to 1024**.

7-3 Fine Tuning the Contrast and Brightness using the SMPTE Pattern

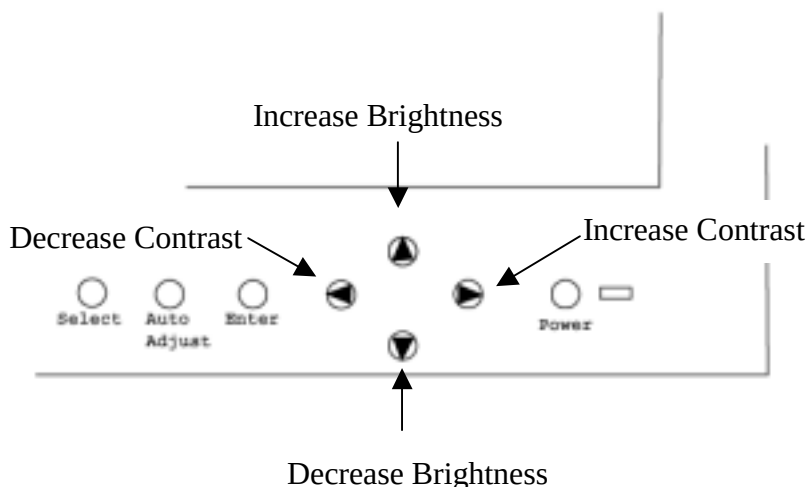
In this section you will adjust the Contrast and brightness controls of the LCD Monitor to optimize the SMPTE Test Pattern for the Window and Level controls.

Minimal adjustment should be necessary at this time. However, it is important that you "fine tune" the Contrast and Brightness level using an industry Standard Test pattern.

This is an iterative process by adjusting the contrast, then brightness, and then back to contrast.

Note

To reduce visible tearing or smearing of the pattern, LOWER the contrast.



EIZO LCD MONITOR CONTROLS
ILLUSTRATION 7-4

1. Use the **Control** buttons on the front of the monitor, to adjust the Brightness and Contrast .
2. Look at the Pattern. If the 5% and 95% patches are not visible.(item 6, Illustration 7-3), adjust the brightness control of the display monitor until you can just barely see them. You may also need to re-adjust the contrast, if tearing or smearing of the pattern occurs. (items 1 and 2, and 5 through 8, Illustration 7-3).Continue adjusting both brightness and contrast until the image is as crisp and clear as you can make it, and the 5% and 95% patches are just barely visible.
3. When satisfied with the display, exit the menu to save the adjusted Contrast/Brightness settings.

7-4 Locking the Screen Manager Settings

The Eizo L660 is equipped with an adjustment-locking feature. However, it does not prevent the Operator from adjusting Contrast and Brightness settings.

To lock access to the Screen Manager:

1. Turn off the LCD Display with the front power switch.
2. Press and HOLD the Auto Adjustment button, at the same time push the Power button to turn the LCD Display ON. This will disable access to the Screen Manager.

7-5 Unlocking the Screen Manager Settings

To unlock the Screen Manager:

1. Press and HOLD the Auto Adjustment button, at the same time push the Power button to turn the LCD Display OFF. This will enable access to the Screen Manager

Note

These settings will remain in effect even if power is removed or the system is re-booted. Because an operator can tamper with the Brightness and Contrast Adjustments, It is highly recommended that you make all operators aware of the affect they will have on system image quality.

Note

If the operator chooses to deviate from this Brightness and Contrast adjustment, ensure the camera imaging and calibration has NOT been affected.

2. Remove the SMPTE Pattern by pressing the **[ESC]** key or selecting a different Exam from the browser.

8 - CAMERA CALIBRATION

This procedure describes the steps necessary to verify and set the camera parameters. Once the display is re-calibrated, it is essential to re-calibrate the camera before the system is used for filming. Although a qualified GE Service Engineer could perform the steps below, it is recommended that the following procedure be performed with the on-site assistance of the camera vendor field engineer.

8-1 DASM Interpolation Setup

1. Select the **Install** soft key from the Service Desktop. When prompted, login using the password **operator**.
2. Select the **DASM** folder from the top of the Guided Install GUI once it appears.
3. Set the DASM Interpolation method to **linear**.
4. Before exiting the GUI, insert the SaveInfo MOD or new MOD into the drive. Go to the top left corner of the Install GUI and select **<File>** and then **<Save GI Configuration to MOD>**. This process does not create a SaveInfo disk. It just creates a copy of the information already entered in the GUI tabs for use in the next software install.
5. Exit the install GUI by selecting **<File>** then **<Quit>**. If any error conditions still exist, you will be warned that no changes will be made before exiting.

8-2 Camera Imaging Look-up Table

Verify with the camera vendor field engineer that the currently installed camera lookup table is designed to provide perceivably linear contrast for the light box conditions in the customer's viewing area. If not, request that the camera vendor field engineer replace it with the appropriate look-up table.

8-3 Camera Maximum Optical Density

Note that for optimal reviewing, the light box luminance of the diagnostic region of the film should be in the range of 50 to 500 nits. Vary the maximum optical density setting of the camera to compensate for the light box and to meet this value. A good starting position is a maximum density of 2.8.

Note

The final OD settings may be refined by the radiologists performing the image review.

8-4 Camera Contrast

1. With the maximum /minimum optical densities set to compensate for the review area's light box, select a look-up table for your camera that will produce a perceivably linear gray scale for the same light box and the overall ambient light conditions of the viewing area.

Note

The DICOM 3.14 Standard specifies the Barten's curve for linear perception. It is recommended that the manufacturer base perceptual linearity on this curve.

2. Film the SMPTE pattern on a 1-on-16-format display. Verify that the 5% and the 95% levels are visually equivalent. If not, perform a Contrast test with the SMPTE pattern. Select the new contrast setting from the contrast image set. A good value for the Imation DryView is 3. Use the camera's calibration procedure to set the contrast setting. Ensure that the camera maintains a perceivably linear gray scale.

Note

Filming the SMPTE pattern for contrast calibration may be optional for the camera vendor.

3. Ask the technologist to display a clinical image and set window and level controls for the desired appearance. A good image to start with is a sagittal or axial head image.
4. Capture the image on the keypad or host control interface.
5. Print a Contrast Test film. Ask the technologist to select the image that best matches the displayed image on the monitor.
6. Observe the image number below the selected image and set the Contrast control to this value.

8-5 Anatomical Filming

This portion of the procedure requires the technologist to verify the camera settings with true anatomical images.

1. Film representative anatomical images to confirm the settings. The image set should include T1 and T2 head images, joint images and c-spines.
2. Observe the accuracy of the low-tones, mid-tones and high-tones. If a filmed image is found to not be equivalent then re-calibrate the camera based on the customer's evaluation.

9 - TROUBLESHOOTING GUIDE

The following section provides suggestions for troubleshooting the LCD monitor:

9-1 Explanation of Default Gamma Settings

The Default gamma setting is different for different software versions. The Table below indicates the suggested nominal setting. The Software Default Settings are set at boot up. When a Look-Up-Table is installed, these default settings are ignored. If the “setgamma” command is used, the “colorPix.cfg file and the boot script (Startmon) is updated, with the new single value.

TABLE 9-1
DEFAULT GAMMA SETTINGS

LCD Monitor	Nominal Gamma value
Eizo L660	0.76

As the Gamma setting gets lower, the display will become dimmer, (less contrast).

9-2 Complaint: Poor SNR, Dynamic Range Lack of Grayscale, Blotchy images

- It is important to note that the software default gamma setting of 1.5 causes extremely poor image quality on both the LCD and CRT. Customers complain of Ghosting or Poor SNR or lack of dynamic range in the grayscale. Before troubleshooting any system problem related to these symptoms check the Monitor Gamma Calibration, or make sure you have the correct Monitor Selected.
- Gamma Calibration tables (LUT's) are unique from one monitor type to another. DO NOT use a gamma table designed for a different monitor. If the gamma table or Monitor Type does not exist, start with a single gamma value of 1.0. It may be necessary to “tweak” this setting for your customer. Use the “**setgamma**” command to do so.

9-3 Complaint: Ghosting or Blurring images or Sheering alphanumeric LCD 's ONLY.

- Ghosting and blurring related to areas of gray in an image. This can be related to system problems but also can be attributed to monitor Frequency Rate (Frame and Swap rates). ALL LCD's should be set to 60Hz.

Checking the Frame Rate and Swap Rate Setting:

- From the Desktop click on the **C-shell button**
- Switch to root owner. Type: **su root** [enter]; (Enter password)
- Type: **toolchest&** [enter]
- Click on **Find → Control Panels**
- Double click on Display Icon** to open Display tools.
- Verify: **Active Size (Pixels): = 1280x1024 and FRAME RATE and SWAP RATE =60.00 Hz** (Refer to Illustration below:



FRAME RATE AND SWAP RATE
ILLUSTRATION 9-1

7. If it is the Frame Rate and Swap Rate are not set to 60Hz:
8. Select **1280X1024_60** from the left menu.
9. Click on **"Load"** button.
10. Click on **OK** when the Pop-up box asks you if you want to Load the setting now.
11. Click on **OK** when the pop-up box asks you if you want to keep the Format.
12. Click on **OK** when the pop-up box asks you if you want to make this setting the Power-On Default.
13. Close the Graphics Backend Window: Click on **File** → **Exit**
14. Close the Control Panel Window: Click on **Page** → **Exit**
15. Close Toolchest Window: **Right Click and hold on the word Toolchest** at the top of the toolchest window.
16. Drag the mouse to **Close**
17. In the C-shell type: **exit** [enter] to exit root
18. Type: **exit** [enter] to exit the C-shell.
19. Reboot the system.

9-4 Camera Film has changed after Monitor replacement

Note:

If there is no remote camera in the list, you will not be able access the Film Composer Menu.

- Make sure that the Monitor and Camera Calibration procedures have been performed completely and the correct Monitor Type has been selected and installed.
- Then try setting the camera to **"Replicate"** instead of linear mode, or vice versa.
- For Kodak Cameras try setting to **"Bi-linear"** mode.
- For Other cameras try setting **"Replicate"** mode.

Procedure:

1. Open the Image Browser
2. Open Film Composer
3. Select **configure**
4. Select remote printer being used by the site.
5. Select **Update**
6. Under magnification Type:
Select **"Replicate"** or **"Bi-Linear"** option. (Kodak try "Bi-linear")
Select **OK**
7. Select **Done**
8. Select **Quit**
9. Select **OK**
10. Return to original desktop.
11. Check film and adjust camera if necessary

9-5 No Picture

- For Eizo monitor, push the Select button to insure the wrong port is not selected.
- The signal cable should be completely connected to the display card/computer.
- The display card should be completely seated in its slot.
- Power Switch and computer power switch should be in the ON position.
- Check the signal cable connector for bent or pushed-in pins.

9-6 Image Persistence

- Image persistence is when a "ghost" of an image remains on the screen even after the monitor has been turned off. Unlike CRT monitors, the LCD monitor's image persistence is not permanent.

Note

It is still recommended that a screen saver be used whenever the screen is idle.

9-7 Image is unstable, unfocused, or swimming is apparent

- Signal cable should be completely attached to the computer.
- Check the monitor and your display card with respect to signal timings.
- (LCD's Only) Change the video mode to non-interlace and if possible use a 50 Hz refresh rate.
- Adjust Monitor controls to focus and adjust display. Refer to the vendor manual that came with the LCD Monitor for the operation of this function
- (LCD's Only) If your text is garbled, change the video mode to non-interlace and if possible use a 50 Hz refresh rate.

9-8 LED on monitor is not lit (no green or amber color can be seen)

- Power Switch should be in the ON position and power cord should be connected.
- Make certain the computer is not in a power saving mode.

9-9 Display image is not sized properly

- Push the Auto adjustment button.

9-10 Selected resolution is not displayed properly

- Push the Auto adjustment button.

9-11 Diagnostic Image Quality has been traced to problem with LCD setup.

- Operator has adjusted the Brightness and Contrast of the monitor which has affected Camera/Film imaging. Re-calibration may be necessary. Start with Section 1 of this procedure.

9-12 Areas of white in the image displayed appear slightly blue.

- The Color Mode is set at Mode 2. It must be changed to Mode 1 "default" value. You will find this adjustment in the Screen Manger Menus. Refer to sections A2- A6 in Appendix-A to access and change the Mode. You can also refer to the Eizo User's Manual that came with the monitor.

9-13 Areas of white in the image displayed appear slightly red.

- The Color Mode is set at Mode 3. It must be changed to Mode 1 "default" value. You will find this adjustment in the Screen Manger Menus. Refer to sections A2- A6 in Appendix -A to access and change the Mode. You can also refer to the Eizo User's Manual that came with the monitor.

9-14 Display image ghosts or gray lines in areas of extreme black & White transitions.

- The LCD Refresh Rate is too fast and must be slowed down to allow for the Pixels to react properly. To work around the issue all LCD monitors should be set to Refresh Rate, (Also known as a Line Rate or Frame and Swap Rate) of 60Hz. . The trade-off will be: Better Black & White Images at the sacrifice of color smoothness. Letters such as "V" and "W" will appear more jagged.

•

9-15 Display image Blurry or Blotchy images Described as poor grayscale.

LCD's are highly dependent on gamma correction. Perform the gamma correction procedure to solve this problem. See Section 5 of this procedure.

Note: The off-the-shelf LCD's on Signa are optimized for PC use from the factory, NOT diagnostic Imaging. The LCD Monitors are NOT "Plug and Play" with regards to the Signa Applications Software and SGI Computer Video system. This requires that a complete monitor Setup and calibration be performed at:

- Installation
- During a replacement of a failed monitor.
- Any time Software is reloaded.

9-16 Customer complaints of poor grayscale, Window or level, lack of definition.

LCD technology uses three color film plates and a backlight to display different shades of color and gray. They are highly dependent on gamma correction. The off-the-shelf LCD's on Signa are optimized for PC use from the factory, NOT diagnostic Imaging. Attaching an LCD to the Signa system without calibrating it will cause image quality problems that can be interpreted as system problems, leading to unnecessary:

- Hardware replacement
- Protocol Troubleshooting
- Customer Satisfaction Issues.

It is important to note that LCD Monitors are NOT "Plug and Play" with regards to the Signa Applications Software and SGI Computer video system. This requires that a complete monitor Setup and calibration be performed at:

- Installation
- During a replacement of a failed monitor.
- Any time Software is reloaded.

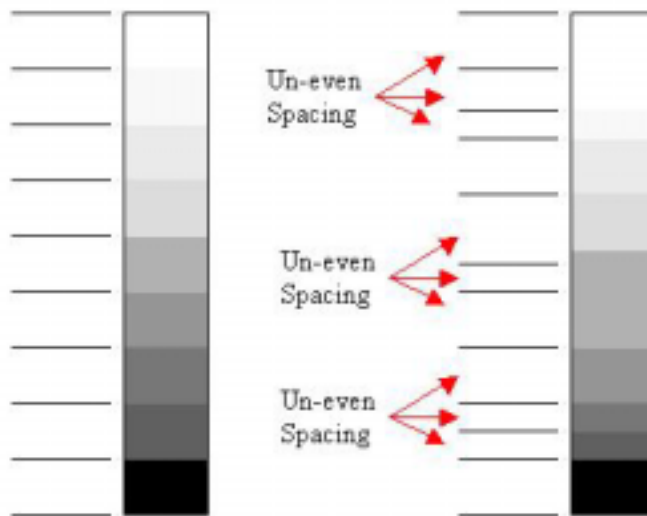
9-17 LCD FAQ's:

- The Monitor calibration information is NOT saved during the SavInfo backup routine or accessible via Healthpage.
- There is a big difference between CRT and LCD monitors. CRT's are much less sensitive to a wrong gamma setting.

- Gamma Tables (Look-Up-Tables or “LUT’s”) are highly specific to the LCD model and should only be applied to the correct model they were designed for.
- Gamma Tables consist of thousands of luminescent data points that are used by the electronic circuitry to generate uniform transitions from one color or grayscale level to the next.
- All recent LCD’s used in Signa have a “best fit - Single gamma value” as well as a complete Gamma Table to maximize flexibility when setting up the monitor.
- It is recommendd that the Gamma Table always be used as a default.

9-18 What is Gamma?

Simply stated gamma normalizes the grayscale so each level is unifor in size and shape across the entire display:



GRAYSCALE COMPARISON
ILLUSTRATION 9-1

Without getting into complicated equations “Gamma correction” adjust the grayscale or color gradient so each “step” is exactly the same size and shade of color or gray. If the “steps” are not uniform in size and color, the monitor will display the error as poor transitions from one step to the other. This appears to the eye as small ghosts or ragged edges.

9-19 How do I set the correct gamma?

- When the Monitor Type is selected in the Install GUI the correct gamma table (LUT) for that monitor is automatically loaded. The gamma LUT remains active even if a reboot occurs.

9-20 Can I use the gamma command to set gamma?

- NO. This command is no longer an effective way to change gamma permanently. This command should only be used if the need arises to temporarily change to a single gamma value, Overriding any previously installed Gamma LUT.
- In the past, we used to adjust gamma using this command. THIS HAS CHANGED. Using the gamma command on systems with an LCD will most likely CAUSE IMAGE QUALITY PROBLEMS.
- Using a Gamma Table will in most cases produce better display image quality than a single gamma value. A Single gamma value is much less effective on LCD's at providing a uniform grayscale.

9-21 How do I determine if the correct gamma Table or setting is installed?

- Open up the Install GUI. Look at the Display Type Field. To the right of the Option box it will either display a monitor type or a single number.
 - o If a Monitor Type is in this box then a "Gamma Look-Up-Table" for that monitor is installed.
 - o If a "Single Value" is displayed, then the CRT/Custom Option was used to set a Single gamma value.

9-22 When I type "gamma" in a c-shell I see a number. Is that the gamma setting?

- NO. It is simply a "Meaningless number" that no longer holds any value. When a gamma table is installed, (By simply selecting the monitor Type in the Install GUI) This value is meaningless. It is a left over from earlier versions of software.

9-23 When I type "gamma 1.05" in a c-shell, the LCD changes in appearance. Why?

- This is complicated because IRIX Operating System Commands such as "gamma" take priority over the Signa Applications Software. The Install GUI Gamma Tools are written in IRIX Script and have less priority. Since the O/S gamma command was invoked with a value, that value will now OVERRIDE the Applications software. The Single Value of 1.05 temporarily overrides the gamma LUT until a reboot occurs. (Upon reboot, the Gamma LUT resumes control).

9-24 A gamma value of 1.05 is what we always used in the past can we still use it?

- The gamma value of 1.05 is no longer used. All LCD and CRT Monitors now use .95 which was determined to be the best default "Single value".
- DO NOT USE THIS COMMAND ANY MORE. The process HAS CHANGED. Using the gamma command on systems with an LCD will most likely CAUSE IMAGE QUALITY PROBLEMS and filming problems.
- Using a Gamma Table rather than a single gamma value, (Such as 1.05) will in most cases produce much less IQ problems than a single gamma value.
- A Single gamma value is much less effective on LCD's ,at providing a uniform grayscale.

9-25 My customer does not like the gamma setting that the Gamma Table provides.

- By all means change it! use the Install GUI, under Display Type" Select "CRT/Custom" and type in a "Single Gamma Value" you and the Customer agree on is satisfactory. A reboot will make the Single setting permanent.
- There is a Single optimized "Nominal" Gamma Value for each monitor type as well as a gamma table. Consult the Troubleshooting sections within the proper monitor calibration procedures to find the "Nominal" value for the monitor type.
- Note: Using a Single value affects the camera to a much higher degree than a Gamma Table does, so be prepared to adjust the camera as well.

9-26 Single value optimization technique.

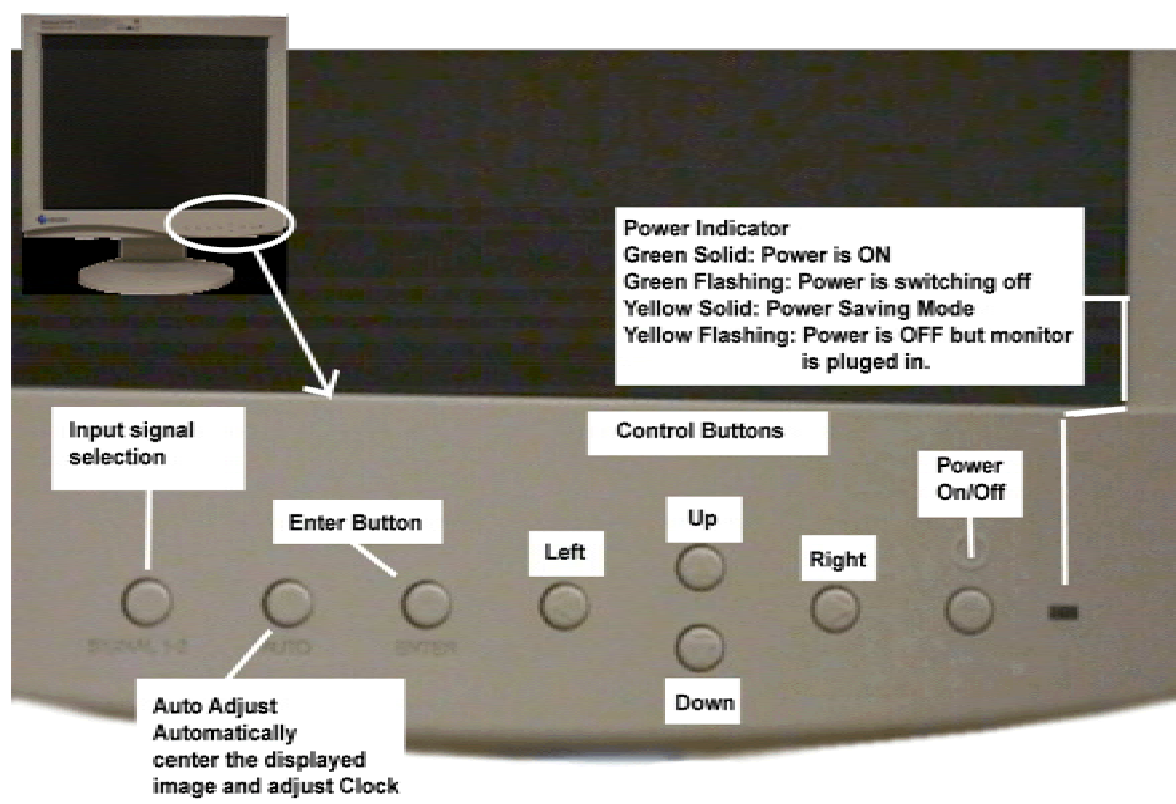
- To minimize the time needed to find a Single gamma value that is acceptable by your customer, you can still temporarily use the IRIX O/S gamma command in a c-Shell to "Tweak" a single gamma value.
 - o Typing "gamma *n.nn*" (where *n.nn* is the nominal single gamma value for the monitor type. It will generate an immediate change to the monitor. (Increasing the value will increase brightness and contrast together. Decreasing it will have the opposite affect).
 - o Continue to adjust the gamma setting, checking images and films until you arrive at a setting you like. Note the final value.
 - o **DON'T FORGET THIS STEP: To make the value permanent** Open the Install GUI and under Display Type" Select "CRT/Custom" and type in the new "Single Gamma Value" .
 - o Reboot the system and this will now be the permanent setting.
 - o Note: If you open a c-Shell and type "gamma" it may not match what you installed in the Installation GUI. This is normal.

A- APPENDIX A

A-1 Focus, Screen position, Clock Rate and Phase Adjustments.

The Auto Adjust procedure done in Section 4 takes care of these for you. All other values in the Screen Manager should stay at the default values except Contrast and Display.

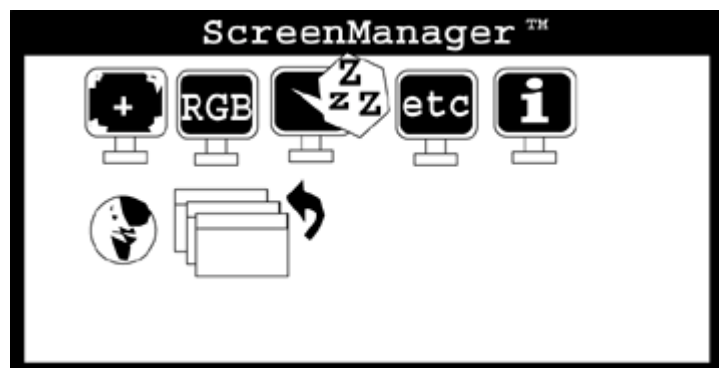
Use this appendix only if you see problems with Focus, screen position relative to monitor mask or phase problems (horizontal bars sweeping through the LCD). Or any vertical problems.



DISPLAY OPERATION
ILLUSTRATION A-1

1. Open the Screen Manager by pressing the **Enter button**.
2. Enter the desired sub-menu by selecting the Icon pressing the Arrows to the desired section.

Screen Manager consists of a main menu and six sub menus.



SCREEN MANAGER
ILLUSTRATION A-2

Table A-1 gives a brief description of each menu item.

TABLE A-1
SCREEN MANAGER ICON DEFINITIONS

Icon	Function	Comments
	Screen	The screen function is used to manually adjust frequency, image position, resolution, contrast and brightness. Only contract and brightness are adjusted to 80 percent for Signa applications. Other features are adjusted using the monitors auto adjust button on the front of the monitor.
	Color	This selections if used to make fine adjustments to the display color of the monitor. These adjustments should be left at the default for Signa applications
	Power Manager	Used to setup the power management features of the monitor. These adjustments should be left at the default for Signa applications.
	Others	Allows adjustments to screen size, border intensity, input priority, power down timer, monitor beeps and if necessary allows resetting the monitor to factory setting. These adjustments should be left at the default for Signa applications.
	Information	This menu shows the current ScreenManager settings. No adjustments are made here.
	Language	Used to set the language the monitor menu will display in. This only effects the ScreenManager menus and has no effect on display data and text being sent from the host computer.
	Exit	Click this Icon to exit ScreenManager.

A-2 Manual Adjustments of Controls.

- Adjust Contrast and Brightness → 
- Image Adjust → 
- Clock Adjustment → 
- Phase Adjustment → 

1. The Screen Manager Window will disappear in 45 seconds unless you select a sub-menu using the appropriate arrow button. If necessary, push the **Enter** button on the monitor to display again.



2. Select the **Screen** menu.

3. If you observe vertical bars or non-linearity's in the pattern, select the **Clock**



adjustment function to reduce the vertical distortion on the monitor. Use the Right and Left Arrow buttons to eliminate this problem

4. If you observe horizontal bars or non-linearity's in the pattern, select the **Phase**



adjustment function to reduce the vertical distortion on the monitor. Use the Right and Left Arrow buttons to eliminate this problem

5. Press the Enter button to memorize both the Clock and the Phase setting.

6. If the screen position needs to be aligned, select the **Position** icon  and **push enter button**. Adjust the image position using the controls to align the screen.

7. Adjustment markers will appear. Use the arrow buttons to align the screen to the adjustment markers.

8. Press the **Enter button** to memorize both the Position setting.

9. Press the **Resolution button** and select the desired resolution. **1280 X 1024**

Note

75 represents 75Hz. This is the maximum frequency rate that this setting can use. The Display Controller of the system will be should set at 50Hz, which is the the optimum setting to maximize image display. (Refer to Section 6)

10. Press enter button to memorize the setting.

11. Select **Brightness/Contrast**  icon.

12. Push the Auto Adjustment button and the COntrast and Brightness adjustment will be automatically adjusted. The Screen blanks momentarily as it adjusts to the maximum contrast value.

13. Adjust the brightness to the desired level.

14. Push the **enter button** to save the adjusted settings.

15. Color Adjustment should be set to **MODE 1**

16. Gain Adjustment. Before setting the Gain adjustment always start by using the "**Reset**" button to set the gains to the default value.

a. set all gains at 100%.

17. Power-Saving Setup- This should stay at the default setting.

18. All other setting should remain in the default position.

19. Use the **return button** to go back to the Main Screen Manager Window.

20. To SAVE all values entered and EXIT, Select the **Exit Icon** or press the enter button twice.

Note

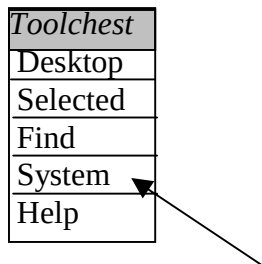
Adjustment information is not lost when the power is removed from the monitor.
Front Panel Controls.

The functions of the controls on the front of the monitor are shown in Illustration A-1

A-3 Adjusting Focus Using LX Calibration Tools From Toolchest (Alternate)

This section is normally not needed. However, if you cannot adjust the monitor satisfactorily, with the front panel buttons and ScreenManager, you may want to try the adjustments described in this section.

1. From the Tools Menu on Signa Open a C-Shell.
2. Type: **[toolchest&]**. A pulldown menu will appear in the upper left-hand portion on the display.



TOOLCHEST MENU
ILLUSTRATION A-3

3. Select **[System]** with the Right mouse button.
4. Select **[Confidence Tests]** from the extended menu. A "confidence test" pop-up will appear after a few seconds.
5. From the "confidence test" pop-up, select the **[Monitor Icon]**. A monitor calibration menu will appear on the screen.
6. Select **[Focus]** from the "Test Options" menu by using the **Tab** key.
7. Put the mouse pointer on the "Test Options" Menu. Click the **Middle mouse button** to toggle (Hide) the "Test Options" menu.
8. Push the auto-adjustment button on the front of the monitor. This should setup Focus, Horizontal, Vertical Centering, sync to the proper Clock Rate and monitor Phase.
9. Observe the test pattern displayed, it should now be centered and in focus. This should be all that is necessary to adjust the display controllers' output to match the LCD monitor.
10. Toggle the "monitor calibration" toolbar back on by selecting the middle mouse. Select the grayscale menu option.

11. Toggle the menu back off by selecting the middle mouse again.

A-4 Setting up LCD Display for Contrast/Brightness. Using LX Calibration Tools From Toolchest (Alternate)

This alternative procedure uses the Monitor Calibration tools and patterns. Steps 1 through 4 should be done while observing the image for crisp lines, overall uniformity of pattern, and to setup the LCD monitor so it is at its brightest level without "Blooming the image or adding distortions". The intent of this section is to achieve the best image appearance you can with the LCD. At the completion of this section you should check the display against industry standard SMPTE pattern. (See Section 7 of this procedure - LCD CALIBRATION FOR OPTIMUM IMAGE VIEWING)

1. Select "**Contrast/Brightness**" button and **push enter**.
2. Adjust brightness to maximum level **100%**.
3. Look at the monitor and adjust the Contrast and Brightness to optimize the test clarity of the test pattern. This is an iterative process between both settings.
4. When satisfied with the displays' contrast and brightness setting is properly achieved, **push the enter button to save the adjusted Contrast/Brightness settings**.
5. Exit the Monitor Calibration by selecting "**Quit**".
6. Select "**File**" on the Confidence Test Window and choose "**Exit**".
7. **Click on the word "Toolchest"** at the top of the menu with the **Right mouse button** and select "**Close**".
8. Exit the C-Shell by typing: **Exit [Enter]**

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
A	July 18, 2001	D. Hofstetter	Initial Version.
0	Nov 8, 2001	D. Hofstetter	Changed Section 5. Added gamma table installation.
1	Dec 14, 2001	D. Hofstetter	Changed Monitor Frequency Rate to 72Hz for all monitor configurations.
2	Mar 18, 2002	Hawthorne	Rearranged Calibration steps
3	Feb 22, 2003	D. Hofstetter	Added correct Look up Table installations for Octane 1 and Octane 2 and Indigo2.
4	Feb 26, 2003	D. Hofstetter	Added Troubleshooting section.
5	May 12, 2003	D. Hofstetter	Added Verification of Frequency Rate Setup. Section 4.
6	June 4, 2003	D. Hofstetter	Streamlined Section 4. Line Rate Setu. Added Troubleshooting solutions.
7	July 25, 2003	D. Hofstetter	Remove old gamma file info and added Troubleshooting info.